Biospectra : Vol. 10(2), Sept., 2015, pp. 5-10

An International Biannual Refereed Journal of Life Sciences

Animal Science

ISSN : 0973-7057

A study of photoperiodic manipulation on the body weight, growth and development of ovarian follicles of *Coturnix-coturnix*

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Received : 12th May, 2015; Revised : 17th July, 2015

Abstract : The live and mature specimens of *Coturnix-coturnix* (female) which is commonly known as (Grey-quail) were purchased from the local bird market in the month of October, 2012. These specimens were acclimatized in laboratory condition and treated in different photoperiodic schedule under observation of 75 days. It is experimentally concluded that birds grew faster and sexually mature early in 22hrsL/2hrsD and 18hrsL/4hrsD, but in short term of photoperiod i.e.; 4hrsL/20hrsD, growth and development of ovary were retarded. The body weight of the birds was also affected in different light regime. It increased in long term of light/dark schedule and decreased in short duration of light. The results clearly indicated that there were a direct relationship between photoperiod and body weight, growth and development of ovarian follicles. In the Natural Day Length (N.D.L.), there were no significant changes recorded.

Keywords : *Coturnix-coturnix*, Ovarian follicles, photoperiodic schedule, light regimes, Natural Day Length (N.D.L.)

INTRODUCTION

Hormonal balance is very essential factor required for proper functioning of the body in animals. Hormonal concentrations vary in different species in the reproductive cycle and also during the different months in the same species. These variations appear to be related to the body weight, metabolic rate and reproductive cycles in variety of birds. It also depends on the hormonal binding capacity of the plasma. The metabolism of estrogen is altered when quails are photo stimulated, castrated or treated with hormonal treatment (Nicholls, *et al.,* 1988)[1].

Day Length plays an important role in the seasonal biology in variety of birds. The reproductive activities of some wild and domestic birds, especially tropical and subtropical regions are guided and regulated by the photoperiod (Wilson and Follett, 1975)[2]. The timing of their migration and their annual reproductive cycles is directly affected due to seasonal changes in day length. (Farner and Lewis, 1971; Farner et *al.,* 1977)[3,4]. The nature of the photoperiodic response and their mechanisms has been experimentally investigated in many avian species in tropical region (Truek, 1978; Device *et al.,* 1982)[5,6]. Administration of corticosterone decreased, the plasma concentration of L.H. in House-Sparrow (Wilson and Follett, 1975; Balthazart and Hirschberg, 1979)[2,7].

In variety of bird species, the effect of long day length was observed on reproductive responses like the photo-induction of gonadal growth, photostimulation, early maturation of gonads; gonadal regression and photo refractoriness (Farner *et al.,* 1983; Bunning, 1973; and Silverin, 1978)[8,9,10].

In some avian species, the circadian rhythm free runs strongly in darkness and near maximal rates of gonodotrophin secretion and gonadal growth occur when birds are exposed to Nanda-Hamner resonance schedules, to single pulses of light when free running in darkness or to T-cycles (Hamner, 1963; Follett *et al.,* 1974; Truek, 1974; Tewari and Kumar, 1981; and Kumar and Tewary, 1982)[11,12,13,14,15].

Yoshimura *et al.,* (2003)[16], suggested and gave an idea for the regulation and photoperiodic response in growth

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and development of ovary in birds. He concluded that light/dark term of photoperiod, acts as a controlling factor in the reproductive cycle of birds.

The aim of the present study is to prove the role of light/dark schedule in the growth and development of ovary in the *Coturnix-coturnix*. The present work also carried out the importance and relationship of photoperiod in experimental birds. The data were also tried to find out, the photoperiodic manipulation in the body weight and ovarian development in the reproductive cycles of birds (Deviche *et al.,* 1982; Urbanski & Follett, 1982)[6,17].

MATERIAL AND METHODS

The experimental birds *Coturnix-coturnix* i.e. female Grey quail were purchased from the fowler in the month of October, 2012 and maintained in an open aviary in an ideal laboratory condition in the Department of Zoology D.A.V. College, Kanpur for 30 days for acclimatization. Just after proper acclimatization on 5th November, 2012, the experimental birds were divided into four groups. Each group having eight female birds, these birds was put into iron wire net cages. The size of experimental cage was 18(L) x 12(B) x 9(H) inches.

The division of experimental groups and duration of experiment was as follows in Fig. No.1

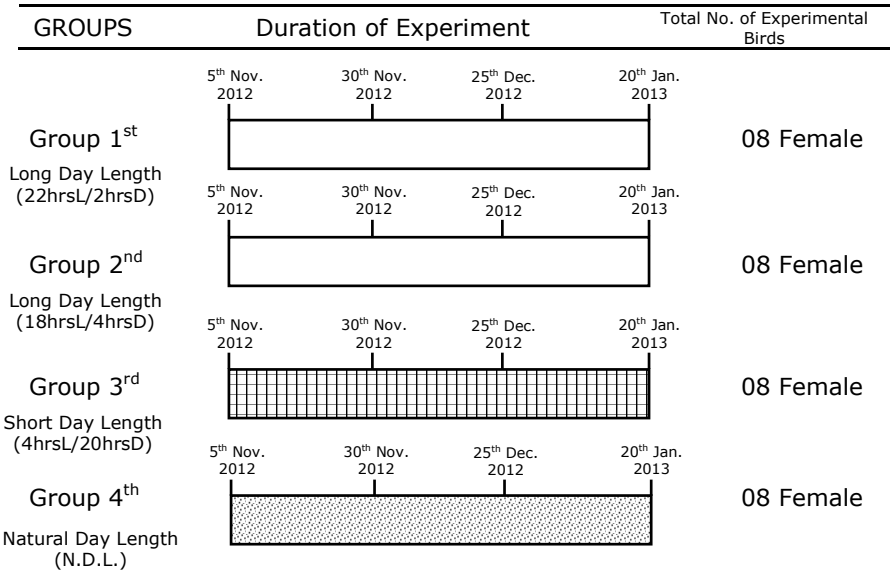


Fig. 1 : Experimental Bird : *Coturnix-coturnix* (Grey quail)
Duration of Experiment : 5th Nov., 2012 to 20th Jan., 2013.

The birds of group 1st to 3rd group were kept in the photoperiodic chamber i.e. empty wooden boxes; the open side of the wooden box was covered with black cloth. The "L" shaped electrical cone pipes (90°) were fitted in the wooden boxes to provide the fresh air to the birds and avoid the suffocation. The light proof wooden boxes contained experimental cages (birds) were illuminated by C.F.L. Automatic timer were used to control light/dark cycles. The last group of birds i.e. 4th group were treated in natural day length condition (N.D.L.) for comparison purpose.

During the course of experiment, the bazra was used as feed along with rice, maize, cocoons, fish powder, salt and egg shells. In drinking water cod liver oil, vitamins and minerals were provided to prevent them against several diseases to all experimental birds.

The duration of experiment under photoperiodic regulation was 75 days. During experimental period on every 25th day, all the birds from each group were operated by using proper anesthesia and area of the largest follicle in the ovary was measured and stitched successfully. The body weights of the experimental birds were also taken at regular time intervals (Fig.1). The data were statistically analyzed for comparison of body weight and growth and development of ovarian follicles in the ovary with response of light/dark cycles.

After the termination of experiment all the birds were open to free from the cage in an ideal natural condition of environment in the forest area.

RESULTS AND DISCUSSION

In the group 1st and 2nd, the mean body weight of the birds increased significantly from 5th Nov., 2012 to 20th Jan., 2013 due to long term of photoperiod and recorded (58.70 grms) to (68.80 grms) and (56.70 grms to 74.50 grms) respectively (Table-I and Fig.2). In the 3rd group of experiment (4hrsL/20hrsD), the initial mean body weight of the birds, 64.90 grms on 5th Nov., 2012. The body weight decreased gradually and recorded 52.60 on 20th Jan, 2013.

In the 4th group of experiment, i.e. natural day length (N.D.L.), the initial mean body weight were fluctuate and recorded 66.00 and 52.63 on 5th Nov., 2012 and 30th Nov., 2012. The body weight once again increased up to 64.90 grms (25th Dec., 2012) and 60.80 grms (20th Jan., 2013) respectively (Table-I & Fig.2).

Table-II and Fig.3 showed, the ovarian development i.e. area of the largest follicle of birds in different groups at regular time intervals. The result clearly indicates that in the group 1st (22hrsL/2hrsD) and group 2nd (18hrsL/4hrsD), the growth and development of area of the ovarian follicles increased due to photoperiodic manipulation but in the 3rd Group i.e. dark phase of experiment (4hrsL/20hrsD), the diameter of the largest follicle decreased significantly and recorded (0.75mm to N.M.) at the termination of the experiment. It is observed that in the last group of experiment i.e. Natural Day Length (N.D.L.), no significant changes were recorded because birds were put in natural condition of light.

In female domestic bird, *Oiseaux-domestiques*, light plays a very important role in the process of reproduction (Sauveur, 1996)[18]. The photoperiodic sensitivity was observed in domestic birds. The perception of light and role of photoperiod were stimulated daily by manipulation of light/dark schedule and were observed in variety of birds (Kumar, 1977)[19]. The hormonal bases of sexual differentiation in Japanese-quail (*Coturnix-coturnix*-Japonica) were treated with testosterone and short term of photoperiod. It was experimentally observed, that it did not masculize females (Elizabeth and Adkins, 2007)[20].

The present results clearly indicate and provide some more evidence, that body weight, growth and development of ovary (area of the largest follicle) depends upon the photoperiodic response of light (Follett and Robinson, 1980; Urbanski and Follett, 1982; Kumar *et. al.,* 2004; Misra, 2009 and Misra, 2010)[21,17,22,23,24].

ACKNOWLEDGEMENT

I would like to express my sincere thanks to Principal, D.A.V. College, Kanpur and Head, Department of Zoology, D.A.V. College, Kanpur for providing me all the required facilities and kind support and co-operation to carry out study.

TABLE – I

Showing mean body weight (in grams) of *Coturnix-coturnix* (Grey Quail)

Months	Groups Body Weight (in grams)			
	1st Long Day Length (22hrsL/ 2hrsD)	2nd Long Day Length (18hrsL/ 4hrsD)	3rd Short Day Length (4hrsL/ 20hrsD)	4th Natural Day Length (NDL)
5th Nov., 2012	58.70 ± 2.41 c	56.70 ± 4.75 N=3	64.90 ± 3.13 b	66.00 ± 2.20 N=2
30th Nov., 2012	62.10 ± 2.20 b	60.60 ± 3.20 N=2	61.50 ± 2.42 c	52.63 ± 1.27
25th Dec., 2012	66.00 ± 2.30 N=2	64.10 ± 0.77 N=3 a	60.87 ± 2.54	64.90 ± 2.10 b
20th Jan., 2013	68.80 ± 1.55 N=2	74.50 ± 0.75 N=2	52.60 ± 1.27	60.80 ± 2.88

Mean ± SE a,b,c differ from respective controls and represents P<0.05, P<0.01, P<0.001 respectively.
When, N=No. of Birds.

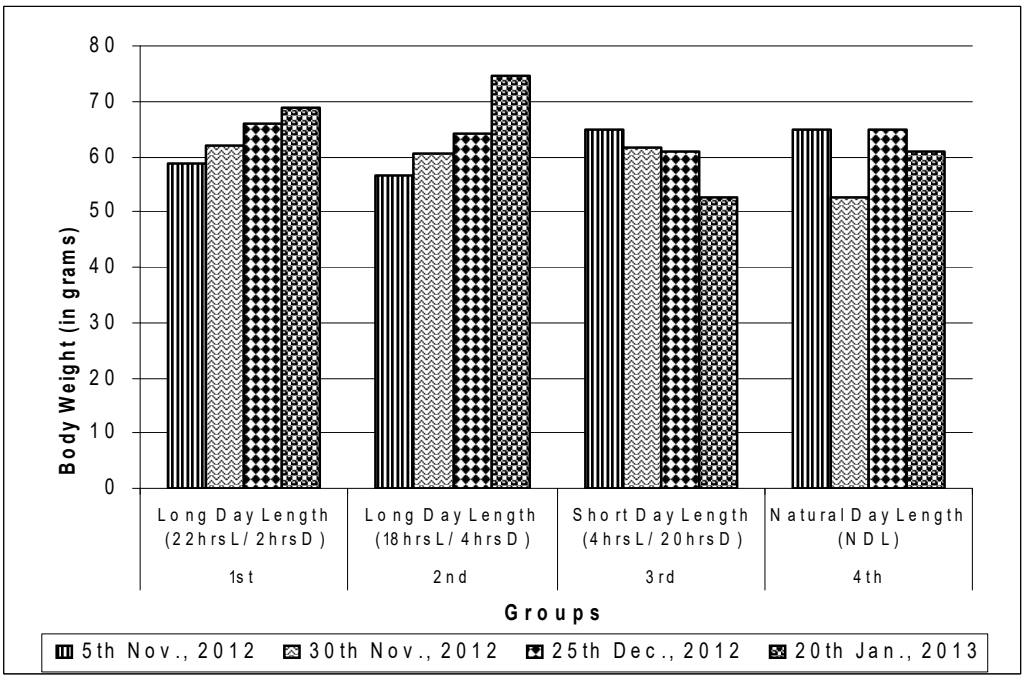


Fig. 2 : Showing the effect of photoperiods on the body weight (in grams) of (*Cotunix-coturnix*).

TABLE – II

Showing the effect of Light/Dark schedule in area of the largest follicles (in mm) of Coturnix-coturnix (Grey Quail)

Months	Groups : Area of the largest follicles (in mm)			
	1st Long Day Length (22hrsL/ 2hrsD)	2nd Long Day Length (18hrsL/ 4hrsD)	3rd Short Day Length (4hrsL/ 20hrsD)	4th Natural Day Length (NDL)
5th Nov., 2012	0.50 ± 00 N=2	NM ± 00 N=2	0.75 ± 0.25 N=2	NM ± 00 N=3
30th Nov., 2012	0.75 ± 0.25 N=2 b	1.00 ± 00 N=3 a	0.66 ± 0.17 N=3	0.62 ± 0.15 N=3
25th Dec., 2012	1.00 ± 00 N=3	1.70 ± 0.30 N=3	0.50 ± 00 N=2	N.M. ± 00 N=2 b
20th Jan., 2013	1.70 ± 0.30 N=3	1.80 ± 0.25 N=2 a	N.M. ± 00 N=2	0.75 ± 0.25 N=2 a

Mean + SE a,b,c differ from respective controls and
$P<0.05$, $P<0.01$, $P<0.001$ respectively.
Where : N= No. of Birds; N.M. = Not Measurable (ovarian follicles are visible in the form of granular spots).

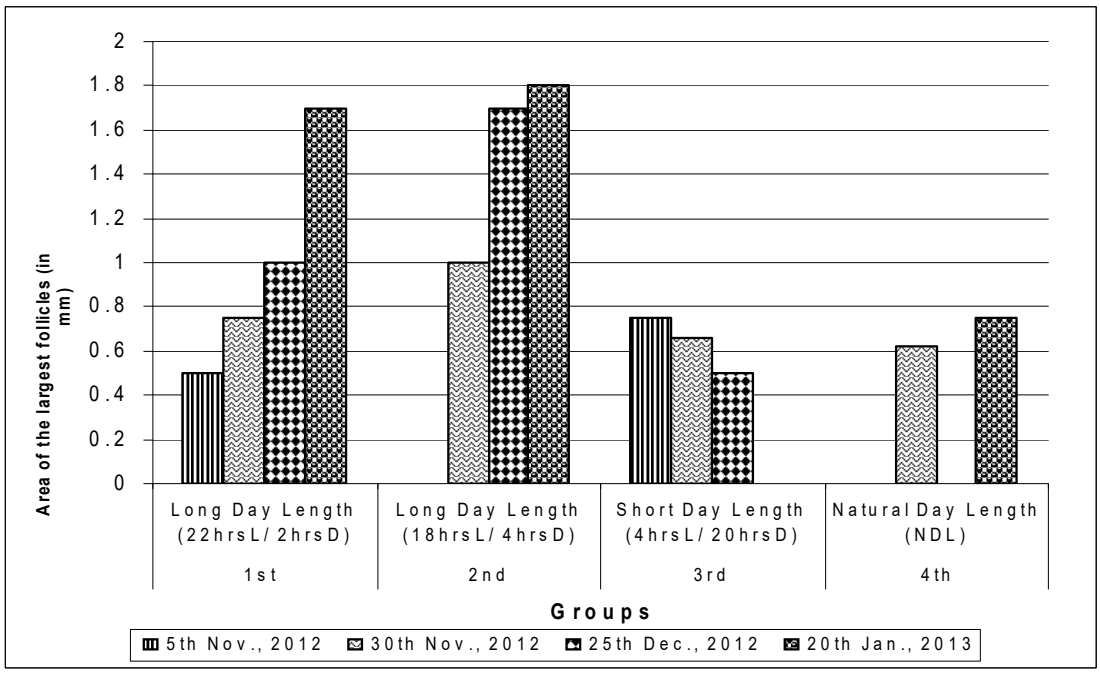


Fig. 3 : Showing the effect of photoperiodism in the ovary (Area of the largest follicles) in mm of *Coturnix-coturnix*.

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